

Weakly Supervised Pneumonia Localization

Machine Learning Mini-Project (UE24CS352A)

Ronit Prasad (PES1UG24CS549) and Vaibhava L (PES1UG24CS512)

1. Problem Statement

Chest X-ray datasets usually come with image-level labels (pneumonia or no pneumonia), but they do not always give precise location labels for every image. In this project we classify chest X-rays as pneumonia or no pneumonia and also try to find the region of the lung that the classifier is using, without using pneumonia bounding boxes as training targets. The region is obtained from the trained classifier using Class Activation Mapping (CAM) and is checked against the ground-truth box using Intersection over Union (IoU).

2. Dataset Details

We use the RSNA Pneumonia Detection Challenge dataset, which provides chest X-rays as DICOM files. After preparation, the working split contains 26,684 unique images, divided as shown in Table 1.

Split	Total images	No Pneumonia	Pneumonia
Train	18,678	14,470	4,208
Validation	5,337	4,134	1,203
Test	2,669	2,068	601

Table 1. Number of images in each split.

The original RSNA labels can contain several bounding-box rows for the same patientId. During data preparation these rows are combined per patient using Target = max, and only the first bounding box is kept. This means evaluation uses one ground-truth box per image, which is a limitation of our implementation. Each image is percentile-normalized, resized from 1024×1024 to 128×128 , and standardized using its own mean and standard deviation.

3. Approach

Classification. A CNN with 10 convolutional layers is trained on the image-level labels. The network is built so that CAM can be applied to it: the last convolutional layer is followed by global average pooling and a single fully connected layer.

Localization. The final convolutional feature maps are combined using the Pneumonia-class weights of the fully connected layer to form a CAM. The CAM is normalized, thresholded, restricted to a conservative chest region, and split into connected components. The largest component gives the predicted box, which is compared with the ground-truth box using IoU. The full pipeline is: DICOM image $\rightarrow$ preprocessing $\rightarrow$ CNN $\rightarrow$ CAM $\rightarrow$ thresholding $\rightarrow$ chest-region constraint $\rightarrow$ connected components $\rightarrow$ predicted box $\rightarrow$ IoU.

Difference from the reference paper. The reference paper uses a U-Net lung segmentation stage and gives both the original and the segmented images to its classifier. We did not train a U-Net because no lung-mask training dataset was available. Our classifier works on the normalized X-ray directly, and a conservative chest-region constraint is applied while processing the CAM instead.

4. Implementation Overview

The network has 10 convolutional layers with 3×3 kernels, padding 1 and ReLU activations, a few 2×2 max-pooling layers, global average pooling, and one final fully connected layer from 128 features to 2 classes. It is trained with Adam (learning rate 0.0001), batch size 32, CrossEntropyLoss, 20 epochs and random seed 42, on 128×128 inputs. This optimizer setting is the same as the one reported in the reference paper. The checkpoint with the best validation accuracy is kept.

Both thresholds were chosen on the validation set only and then fixed before the test set was used. The classification threshold was picked by F1 score: 0.35 gave the best validation F1 of 59.98%. The CAM thresholds 0.20, 0.45 and 0.70 were compared using mean IoU, and 0.20 gave the best validation mean IoU of 0.1050.

The code is written in Python 3.13 using PyTorch, OpenCV, NumPy, Pandas, pydicom, scikit-learn and Matplotlib, and it can run on a CPU or a GPU. It is split into *prepare_data.py* (splitting and annotations), *data.py* (DICOM loading and preprocessing), *model.py* (the CNN), *train.py* (training and checkpoints), *cam_localizer.py* (CAM, thresholding, chest region and connected components), *evaluate.py* and *evaluate_localization.py* (metrics and IoU), *localize.py* (single-image output) and *iou.py* (IoU calculation).

5. Results

Results on the test set, using classification threshold 0.35 and CAM threshold 0.20, are given in Tables 2 to 4.

Classification	Result	Localization	Result
Accuracy	78.57%	Mean IoU	0.1025
Precision	51.85%	Median IoU	0.0646
Recall	67.72%	$\text{IoU} \geq 0.3$	7.82%
F1 score	58.73%	$\text{IoU} \geq 0.5$	0.50%

Tables 2 and 3. Classification and localization results on the test set.

Actual \ Predicted	No Pneumonia	Pneumonia
No Pneumonia	1690 (TN)	378 (FP)
Pneumonia	194 (FN)	407 (TP)

Table 4 shows the confusion matrix at threshold 0.35. Localization was evaluated on the 601 positive test images: 194 of them were below the classification threshold and 407 produced CAM boxes.

Table 4. Test confusion matrix.

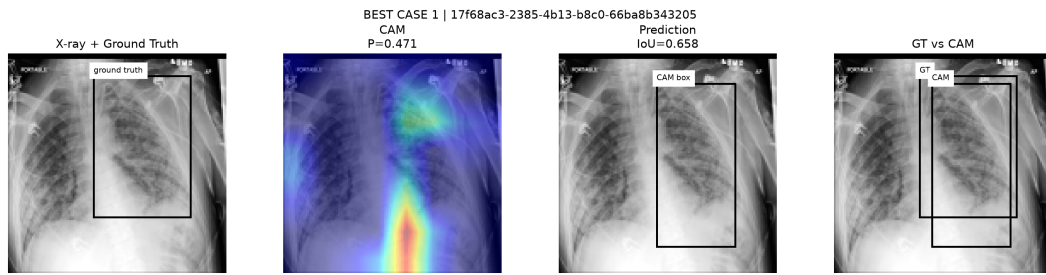


Figure 1. Best case ($\text{IoU} = 0.658$). The predicted CAM region overlaps the ground truth.

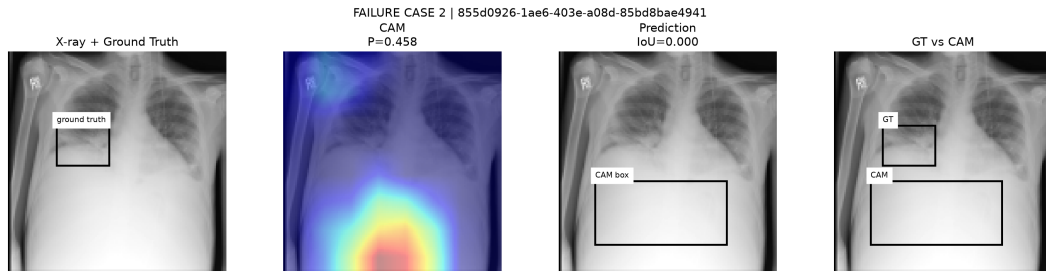


Figure 2. Failure case ($\text{IoU} = 0.000$). The CAM focuses on an incorrect region.

6. Conclusions

The project gives an end-to-end weakly supervised pipeline, from DICOM chest X-rays to image-level classification and CAM-based localization. The classifier reaches 78.57% test accuracy and 67.72% recall at the validation-selected threshold. Localization is much harder: the mean IoU is 0.1025 and only 0.50% of the evaluated positive test images reach $\text{IoU} \geq 0.5$. This shows the gap between recognizing pneumonia and precisely finding where it is.

Limitations: only weak image-level supervision, 128×128 input resolution, approximate CAM localization, classification false negatives, and only the first ground-truth box kept for each image. **Future work:** higher-resolution inputs, transfer learning, a dedicated lung segmentation, better CAM post-processing, multi-region localization, and better handling of images with several ground-truth boxes.